

Antimalarial Drugs

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Malaria

- Most important **parasitic** infection.
- Mosquito borne disease characterized by periodic episodes of fever.
- The periodic episode is due to periodic rupture of red cells with release of merozoites.

Causative factor

- Malaria is caused by protozoa of the genus *Plasmodium*
- Four species of *Plasmodium* :
 1. *Plasmodium vivax*
 2. *P. Ovale*
 3. *Plasmodium malariae*
 4. *Plasmodium falciparum*

Mode of transmission

- *Plasmodium* is most commonly transmitted through **the bite of a female Anopheles mosquito**.
- Malaria can also be transmitted by
 - blood transfusion
 - vertically from mother to fetus across the placenta

Pattern of malaria

- *P. falciparum* Malignant tertian No relapse
 malaria
- *P. vivax* Benign tertian Relapse
 malaria
- *P. Ovale* Benign tertian Relapse
 malaria
- *P. Malariae* Quartan malaria No relapse

Antimalarial drugs

- Therapeutic classification

1. **Causal prophylactics (primary tissue schizonticides)** – destroy parasite in liver cells and prevent invasion of erythrocytes

Primaquine

Proguanil

Pyrimethamine

Mefloquine

2. **Blood schizonticides** – destroy parasites in RBCs and terminate clinical attacks of malaria.

Chloroquine

Pyrimethamine

Halofantrine

Atovaquone

Artemisinin and derivatives

Mefloquine

3. **Tissue schizonticides** – drugs eliminating tissue schizonts or latent hypnozoites (hepatic) and prevents relapse.

Primaquine

Tafenoquine

4. **Gametocidal drugs:** destroy gametocytes and prevent transmission.

Primaquine

Chloroquine

Quinine

Chemical classification

1. **4- aminoquinolines:** Chloroquine ,
amodiaquine
2. **8- aminoquinolines:** Primaquine,
tafenoquine
3. **Quinoline methanols:** quinine, quinidine,
mefloquine
4. **Sesquiterpene lactones:** Artimisinin,
artesunate, artemether
5. **Folate antagonist:** Proguanil, pyrimethamine

6. **Phenanthrene methanol:** Halofantrine, lumefantrine
7. **Naphthaquinone :** Atovaquone
8. **Antibiotics:** Tetracycline, doxycycline, ciprofloxacin, clindamycin, azithromycin

Chloroquine

- A synthetic 4- aminoquinolines
- **Highly effective** blood schizonticide with activity against all species of plasmodia
- **Rapidly acting-** patients become afebrile in 24-48 hours
- Also destroys gametocytes of *P. vivax*, *P. ovalae* and *P. malariae*.
- Safe in **pregnancy**

Pharmacokinetics of Chloroquine

- Route: oral, parenteral
- Absorption: rapid, complete from GIT
- Distribution: **large volume** of distribution, extensive tissue binding (**heart**, kidney, liver, spleen, **retina**, cornea)
- Metabolism : liver (metabolite have antimalarial activity)
- Excretion: urine
- Plasma half life: 3-5 days

Antimalarial actions

- Malarial parasite digest the host haemoglobin (which is their source of amino acids)



Transport it into their acidic food vacuoles



A toxic product “**haeme**” is formed



haeme polymerase

This **haeme** is converted to nontoxic **haemozoin**, a malarial pigment

Antimalarial actions.....

Chloroquine



Enters in erythrocytes



Taken up by the parasite by specific uptake mechanism



Enters in acidic food vacuole of the parasite



Inhibits heme polymerase

Accumulation of toxic heme



Parasite membrane lysis



Death of parasite

Other actions of Chloroquine

- Chloroquine has activity against *Giardia lamblia* and *Entamoeba histolytica*. It attains high concentration in the liver and is useful in **hepatic amoebiasis**.
- Chloroquine has **local anesthetic effect**.
- It has also **anti-inflammatory** properties because of which it is used in **rheumatoid arthritis** as a disease modifying antirheumatic agent.

Indication/use

1. Malaria- highly effective against sensitive strains of all 4 species.
2. Extraintestinal amoebiasis
3. Rheumatoid arthritis
4. Lepra reactions
5. Discoid lupus erythematosus
6. Infectious mononeucleosis.

Adverse effects of Chloroquine

1. Nausea, vomiting- if severe ,before 30 min antiemetics should be given
2. Pruritus, skin rashes
3. Headache, visual disturbances
4. **IV Chloroquine-** causes hypotension,
widening of QRS complex
arrhythmia
respiratory and cardiac arrest
(hence parenteral **Chloroquine is avoided**)

Adverse effects cont.....

5. High dose— cardiomyopathy, peripheral neuropathy, ototoxicity, convulsions
6. Long term suppressive therapy- blurring of vision, confusion, bleaching of hair, rarely blood dyscrasias
7. Prolonged treatment with high dose- **irreversible retinopathy** and can result in **blindness** (as chloroquine accumulates in retina).

Contraindications of Chloroquine

1. Psoriasis and porphyria
2. Hepatic, gastrointestinal or neurological disorders
3. Myopathy
4. Retinal disease (long time administration in high doses require regular neurological and eye examination).

Precautions/drug interactions

1. Parenteral administration should be avoided but when required should be given as a slow infusion.
2. Chloroquine + gold/d-penicillamine → severe dermatitis
3. Chloroquine + quinine/mefloquine → therapeutic failure

(they compete for accumulation in the parasite resulting therapeutic failure)

Following parenteral administration, toxic conc. may be attained in plasma in a very short time

- 4. Chloroquine + mefloquine —————> increase the risk of seizure
- 5. Chloroquine + halofantrine —————> increase the risk of arrhythmia

hence, **a gap of at least 12 hours** should be given, if pts. have to be switched over from chloroquine to quinine/ mefloquine/ halofantrine

- 4. Magnesium containing antacids and kaolin —————> interfere with the absorption chloroquine

Quinine

- An **alkaloid** obtained from bark of cinchona tree
- Destroys **erythrocytic** forms of the parasite similar to chloroquine
- Rapidly acting and is often **effective even in chloroquine-resistant strains of *P. falciparum***
- Also gametocidal for three species of the malarial parasite except *P. falciparum*
- Quinidine, the d-isomer of quinine, can be used in place of quinine

Pharmacokinetics of Quinine

- Rapid well absorption after **oral** administration
- **Widely distributed** in body tissues
- **Long half life** because of increased protein binding (**to alpha 1 acid glycoprotein**)
- Metabolism: liver
- Excretion : urine
- Dose: 600mg TDS for 3-7 days

Antimalarial actions of quinine/quinidine

- Mechanism of action is not exactly known
- May act like chloroquine by inhibiting the enzyme haeme polymerase

Pharmacological effects

- Antimalarial action
- Mild analgesic and antipyretic property
- Myocardial depressant– **IV administration** can cause significant hypotension
- Local anesthetic properties (sodium channel blocker)
- Skeletal muscle relaxant effect
- Stimulates the uterus and is an **abortifacient**

Indication of quinine

1. Malaria : a. uncomplicated falciparum malaria
b. complicated falciparum malaria
c. cerebral malaria
2. Babesiosis- combination of quinine and clindamycin is the first line treatment
3. Muscle cramps: nocturnal muscle cramps
myotonia congenita
4. As anti-epileptic drugs

Babesiosis is a disease infected from the bite of a tick infected with the parasite Babesia....causes flu like syndrome.

Adverse effects of quinine

1. Highly bitter and **gastric irritant**— causes nausea, vomiting and epigastric pain— so, poorly tolerated
2. Hypoglycemia- result in coma. Cause are:
 - a. quinine stimulates the pancreatic beta cells to release insulin
 - b. parasite consumes glucose
 - c. decrease in food intake due to malaria

3. CNS disturbances: cinchonism –

ataxia

nausea

tinnitus

headache

color blindness

dizziness

4. Cardiotoxicity: profound hypotension (**IV rapidly**)

widening of QRS complex

AV block

arrhythmia

5. Allergic reactions:

1. Black water fever (uncommon but fatal): quinine can precipitate acute haemolytic anaemia with renal failure , haemoglobinuria and fever. It is thought to be a hypersensitivity reaction.

2. Skin rashes, urticaria and angioedema

Contraindication of quinine

1. Visual/auditory problems
2. Cardiac arrhythmia
3. Concurrent use with mefloquine
4. Renal insufficiency (must reduce the dose)
5. Should be stopped in case of signs of severe cinchonism, hemolysis and hypersensitivity

Precautions

1. Quinine should **preferably be given orally.**

But in situations where it needs to be given parenterally, **adequate monitoring of cardiac function** is needed.

2. Hypoglycemia should be watched for—adequate glucose supplementation should be given.

Precautions cont.....

3. Quinine should never be combined with mefloquine for the risk of cardiotoxicity.

If the patient has received mefloquine earlier (even up to 20-30 days) and if quinine is needed, it should be used with great caution.

Mefloquine

- A quinoline methanol
- A single dose is **highly effective** against **erythrocytic** forms of the malarial parasite including the multi-drug-resistant (MDR) strains of *P. falciparum*

Pharmacokinetics of Mefloquine

- Well absorbed when given orally
- Not given parenterally --as it causes severe local irritation at the site
- Has a **long $t_{1/2}$** (nearly 20-30 days)- as it undergoes extensive enterohepatic circulation
- Excreted through the **gut**
- **Dose:** 20 mg/kg single dose or in 2 divided dose

Mechanism of action

- Exactly not known but it is thought to act like chloroquine by inhibiting heme polymerase.

Indication

1. Uncomplicated MDR strains of falciparum malaria- mefloquine used with artesunate
2. Chemoprophylaxis of chloroquine resistant falciparum malaria-- mefloquine used with chloroquine

Adverse effects

- **Common-** nausea, vomiting, dizziness, confusion, headache, abdominal pain, sleep disturbances
- CNS effects- disorientation, visual and auditory disturbances, seizure, encephalopathy, psychotic disturbances
- CVS- depression of cardiac conduction resulting in bradycardia and arrhythmias

Contraindication

1. Epilepsy
2. Psychiatric disorder
3. Cardiac arrhythmia and conduction defects
4. Hypersensitivity

Lumefantrine

- Schizonticidal against erythrocytic forms of all plasmodium species including MDR strains of *P. falciparum*
- Actions similar to mefloquine
- Given orally but absorption is erratic
- Food enhances absorption
- Excreted in the stool.

Indication

- Artemether and lumefantrine combination (Coartem) is used in treatment of **MDR falciparum malaria**

(the combination should be given with fatty food to ensure good bioavailability)

Primaquine

- Effective against all forms of the malarial parasite **except erythrocytic** forms.
- **Causal prophylactic:** it destroys the parasite in the liver cells and prevents the invasion of erythrocytes.
- **Hypnozoitocide:** Primaquine destroys hypnozoites (exoerythrocytic forms) in the liver and thereby prevents the relapse of *P.vivax* and *P. ovale* malaria.

- **Gametocide:** destroys the gametocytes of all four species of the malarial parasite, **so interrupting the life cycle and preventing the infection of mosquito.**
- **Mechanism of action:** is not known

Indications of primaquine

1. **Radical cure of *P. vivax* and *P. ovale* malaria:** along with a blood schizonticide.

Dose: 15 mg/day for 14 days

2. **Gametocidal effects:** Primaquine is used for its gametocidal effects in *P. falciparum* malaria--- 45mg single dose.

Radical cure : Rx of both the blood stage and liver stage of malarial parasite... obtained by primaquine that eradicates liver stage of *P. vivax* malaria

3. **Terminal prophylaxis:** After a visit to an endemic area (along with regular chloroquine prophylaxis), a course of primaquine is recommended to destroy the liver forms and gametocytes—15-30 mg for 14 days.
4. **Chemoprophylaxis of falciparum and vivax malaria**
5. **Pneumocystis jiroveci:** In mild to moderate pneumocystis, as an alternative to cotrimoxazole, a combination of clindamycin with primaquine may be used.

Adverse effects

- Primaquine is well tolerated in therapeutic doses. There may be-
 1. Epigastric distress
 2. Agranulocytosis
 3. **Haemolytic anaemia** –in glucose-6-phosphate dehydrogenase deficiency

Contraindications of primaquine

1. Pregnant women (as fetus is relatively G6PD deficient)
2. In glucose-6-phosphate dehydrogenase deficiency
3. Granulocytopenia or methemoglobinaemia

Proguanil

- A biguanide
- A blood schizonticide but also has causal prophylactic activity.
- The onset of action is slow and when used as monotherapy resistance develops rapidly. Hence , it is always used in combination with atovaquone in malaria.

Mechanism of action of Proguanil

- Proguanil, a prodrug, is converted to cycloguanil in the body

- This metabolite is an inhibitor of dihydrofolate reductase in the *Plasmodium*

- Inhibits folate synthesis

Indications of Proguanil

1. Causal prophylaxis
2. Chloroquine resistant falciparum malaria (along with atovaquone)
3. Toxoplasmosis
4. *Pneumocystis carinii* pneumonia

Adverse effects of Proguanil

- A/E are minor including nausea, vomiting, diarrhoea, abdominal pain and rarely haematuria

Atovaquone

- Effective against the erythrocytic forms of plasmodia.
- When combined with proguanil, the activity is synergistic and development of resistance is less common.
- **Malaron=Atovaquone (250mg) + Proguanil (100mg)**

Mechanism of action of Atovaquone

- Atovaquone inhibits mitochondrial electron transport of malarial parasite
- 
- No metabolism of food of malarial parasite
- 
- No energy production
- 
- Subsequently death of malarial parasite

Indications of Atovaquone

1. Treatment of chloroquine resistant falciparum malaria (malarone)
2. Chemoprophylaxis of chloroquine resistant falciparum malaria (malarone)
3. Atovaquone is used to treat *Pneumocystis carinii* pneumonia and toxoplasmosis

Adverse effects of Atovaquone

1. GIT upset--nausea, vomiting, diarrhoea, abdominal pain
2. Seizure, confusion
3. Q-T prolongation

Contraindication of Atovaquone

1. Pregnancy
2. Cardiac arrhythmia

Artemisinin and its derivatives

- Artisunate
- Artemether
- Dihydro-artemisinin

Pharmacokinetics

- Artisunate— water soluble-- oral, IM, IV, rectal
- Artemether-- lipid soluble-- oral, IM, rectal
- Dihydroartemisinin- water soluble-- oral

Mechanism of action of Artemisinins

1. Artemisinin interacts with haem resulting in generation of free radicals
 - free radicals bind to  and subsequently damage to the macromolecules and membrane proteins
 - Death of malarial parasite 
2. It also inhibit calcium ATPase in the parasite.

Artemisinin based combination therapy

- (ACT) is the WHO recommended treatment **for all confirmed cases of falciparum malaria.**
- **Uses:**
 1. In the treatment of resistant/MDR falciparum malaria.
 2. In the treatment of uncomplicated and severe falciparum malaria.

- ACT= Tab. Coartem= Artemether 20mg + Lumefantrine 120mg

- **Advantages of ACT:**

1. Rapid action
2. Better efficacy
3. High cure rates
4. Reduced risk of development of resistance
5. Fewer toxic effects
6. In some cases, shorter duration of treatment

Adverse effects of Artemisinin

1. GIT upset
2. Itching, rashes
3. Anaemia, hemolysis, neutropenia– rare and reversible

Malaria in pregnancy

- P. vivax malaria- chloroquine
- P. falciparum malaria

a. Uncmplicated:

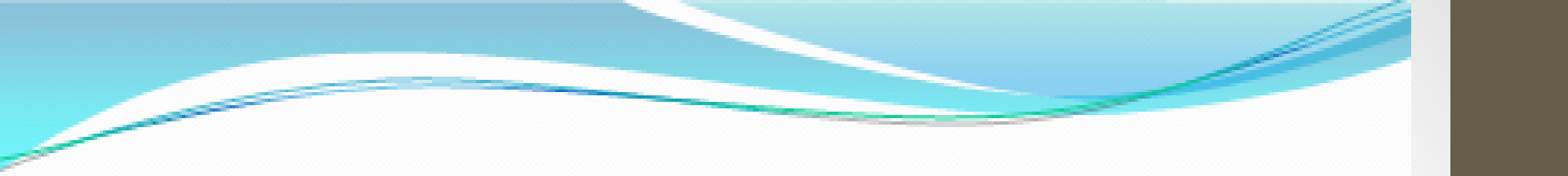
1st trimester- quinine + clindamycin (7 days)

2nd and 3rd trimester- ACT (artemether + lumefantrine)

Alternatives:

1. ACT with mefloquine
2. Quinine

b. Complicated: ACT



That's all for today!

